

Stylizing 3D animation

Final presentation

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Mathieu Senart

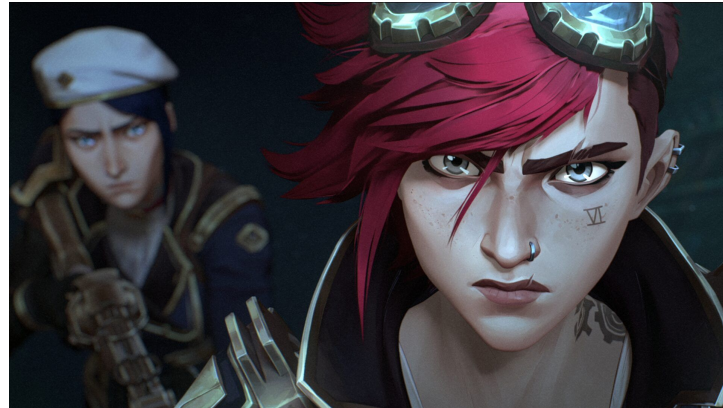
Francisco De Castro Leal Henriques

Supervisors: Amandine LEGRAND, Amal Dev PARAKKAT

Project Overview

Non-photorealistic rendering and the process of stylizing animations nowadays:

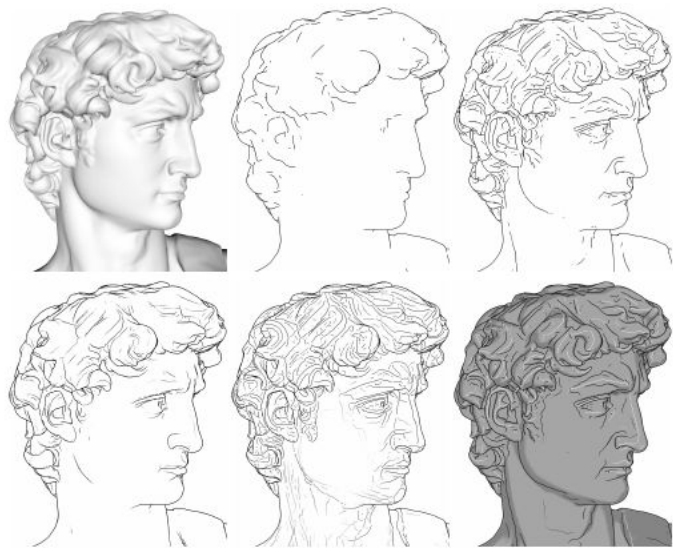
- Post-processing step
- Manual



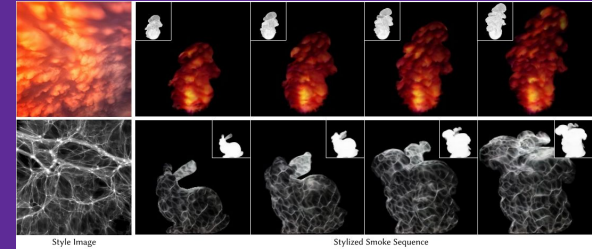
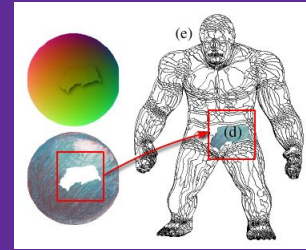
Project Overview

Our objective:

- Investigate the state of the art of animation stylization in earlier steps of the rendering pipeline
- Implement animation stylization techniques (custom shaders, geometry processing) and develop our own solutions



State of the Art Investigation



avior triggered by the skeletal motion, while autom
hape (see giraffe example), localized effects in so
e).

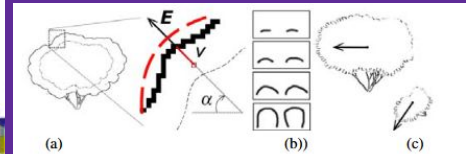
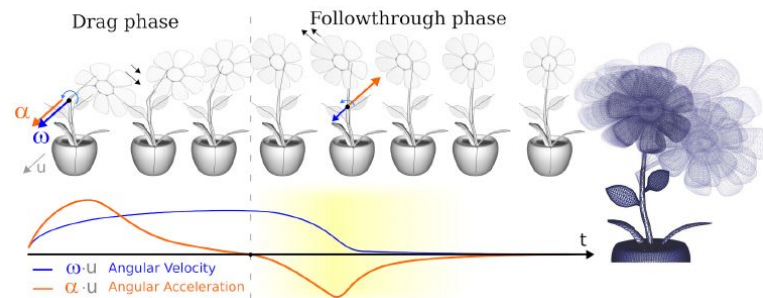
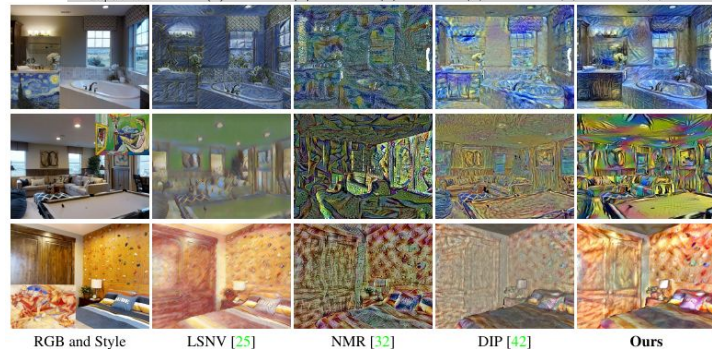
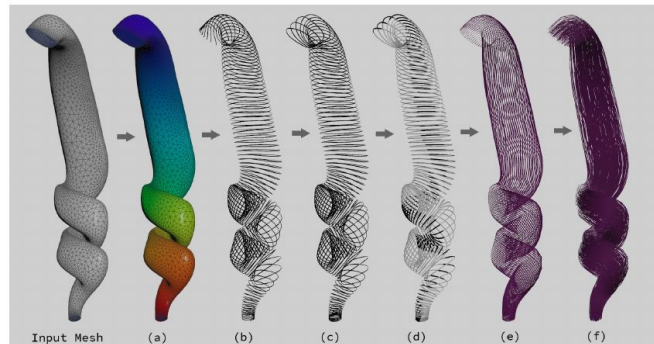
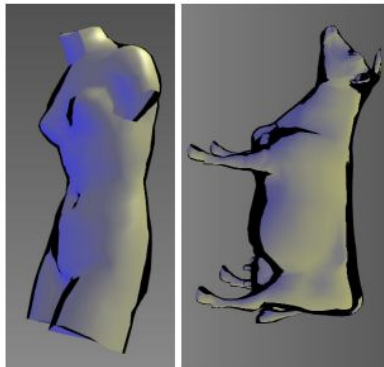


Figure 5: a) Parameterization in image space: 1-pixel contours are expanded inwards (the v coordinate). The parameterization along the silhouette is approximated with the length of the arc described by α on the HLP projection (the dashed line). b) 4-channel SAM (silhouette art map) and c) zooming and rotation behaviour for this SAM: the per-object 2D orientation vector is depicted, which is the reference for α .

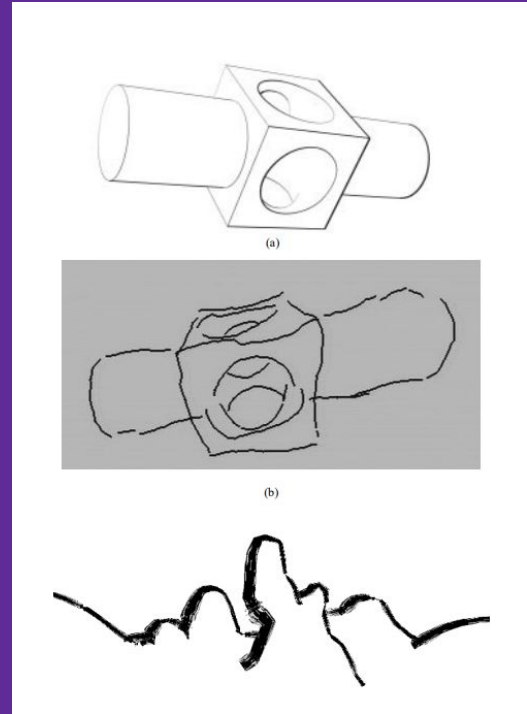
State of the Art Investigation

- Style transfer
- Silhouette detection
- Fluid simulation visualization
- Hatching / Line art
- Toon shading
- Simulation
- Hair stylization
- Motion blur

...

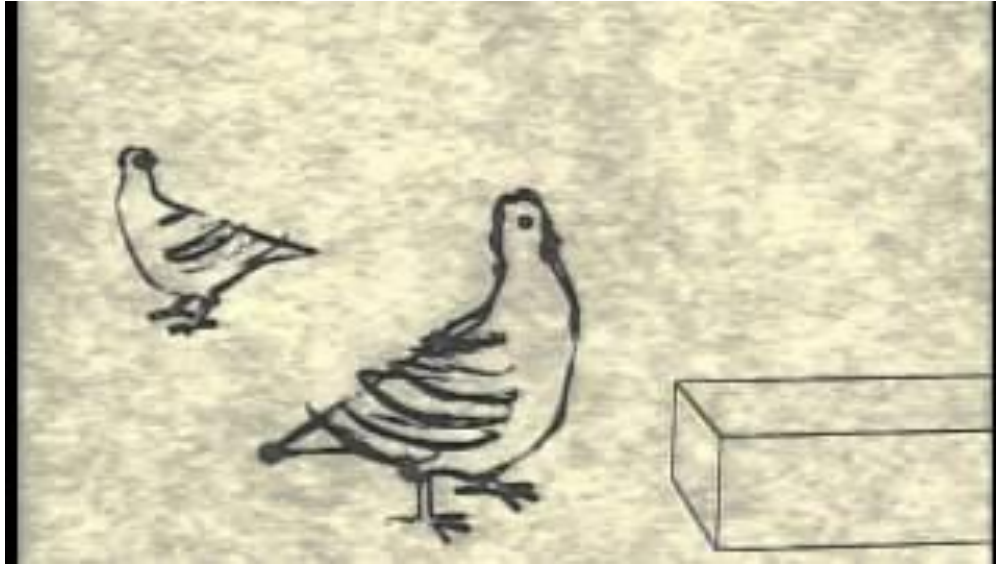


Silhouette stylisation



Source image : MARKOSIAN, L., KOWALSKI, M. A., TRYCHIN, S. J., BOUR-DEV, L. D., GOLDSTEIN, D., AND HUGHES, J. F. 1997. Real-time nonphotorealistic rendering.

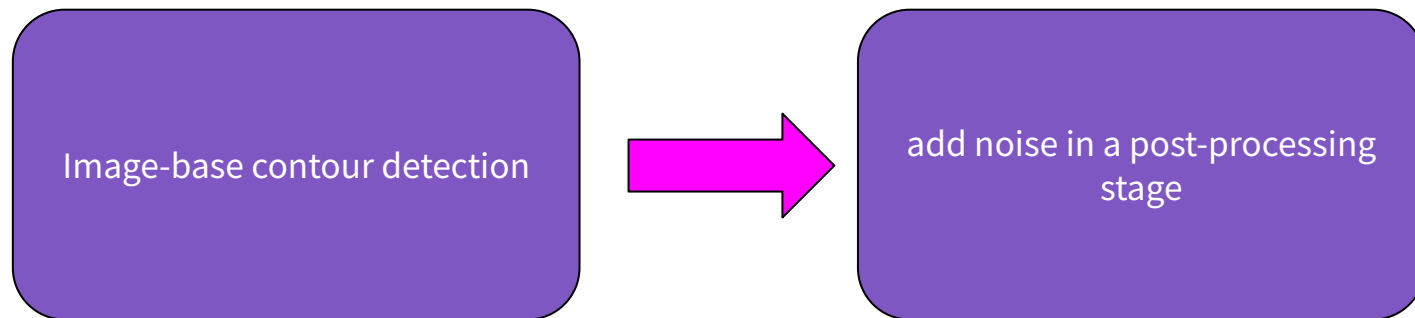
Objective



Loose and Sketchy Animation,
Curtis, 1998

- Create force field from depth buffer
- Use of “particles of ink” to draw sketchy contours
- We will use something simpler and faster

My plan



Render pipeline

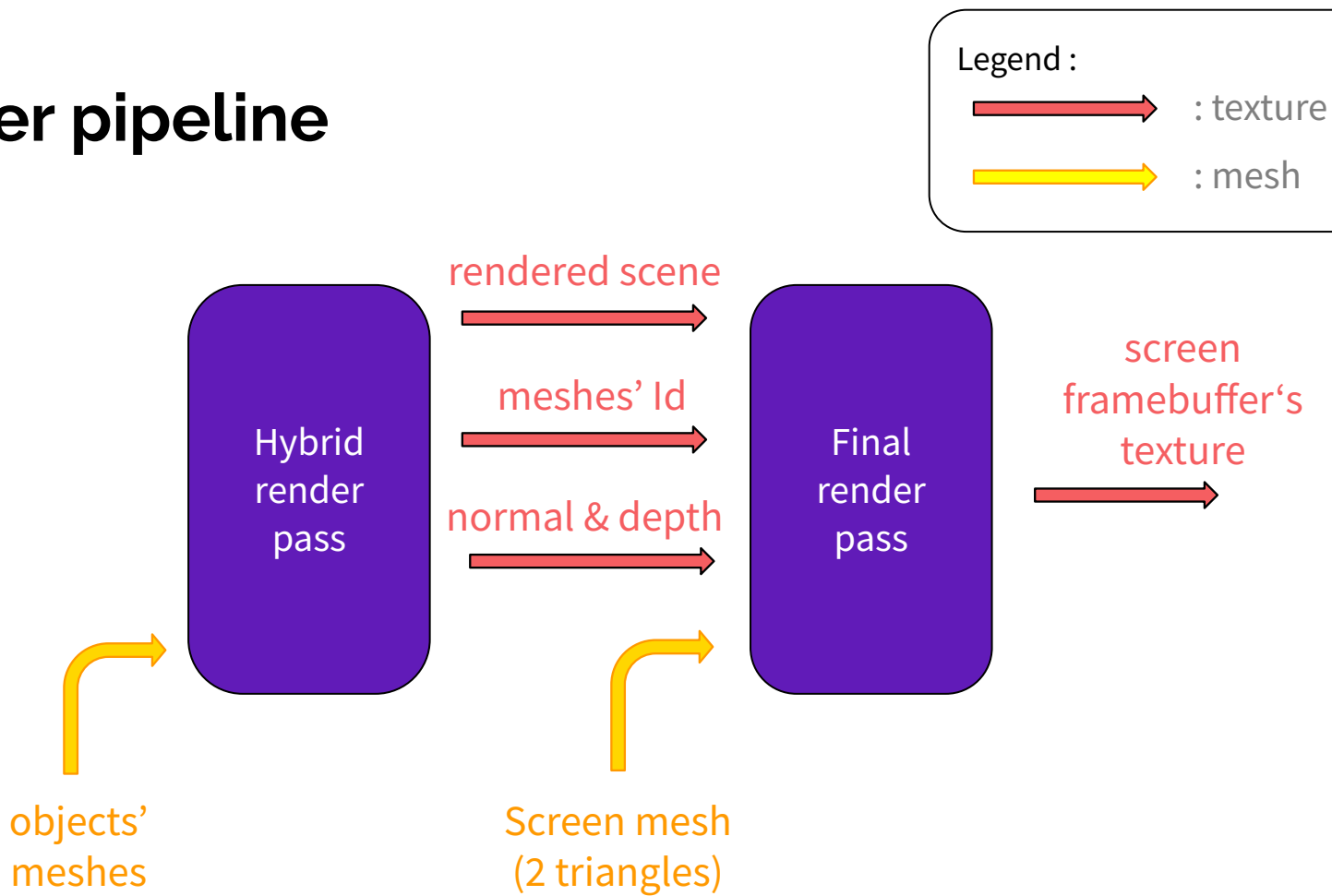


Image-based silhouette detection

Finding discontinuities in z-buffer...

- SAITO, T., AND TAKAHASHI, T. 1990. Comprehensive rendering of 3-d shapes. Computer Graphics, SIGGRAPH 90 Conf. Proc.24(4), 197-206.

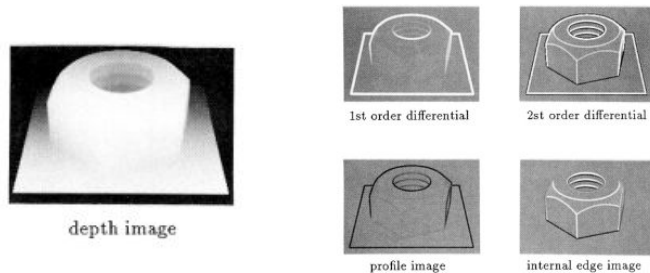
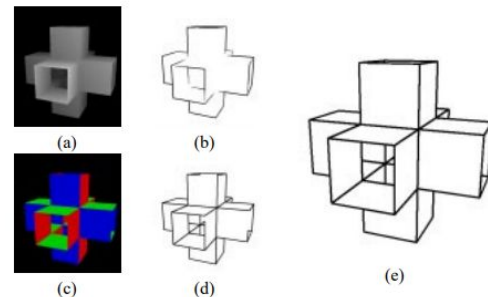


Fig.3 An example of edge drawing.

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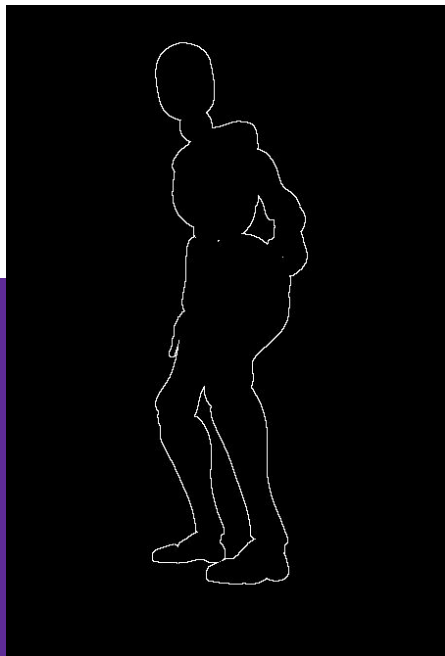
...or in normal buffer

- hertzmann 1999, Introduction to 3D Non-Photorealistic Rendering: Silhouettes and Outlines
- detect more edges, including creases (“plis” in french)

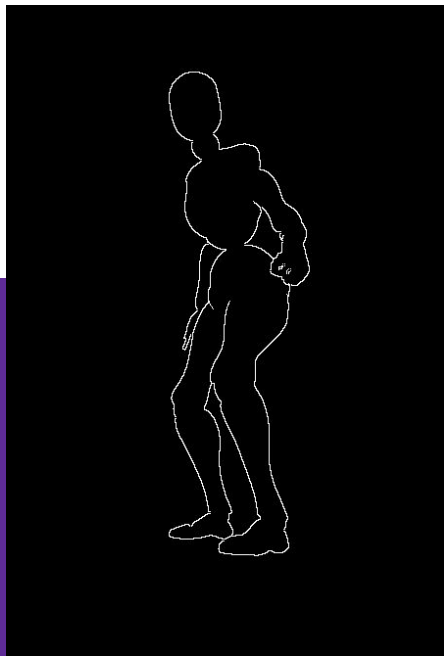


b) using depth buffer
d) using normal buffer
e) combined edge images

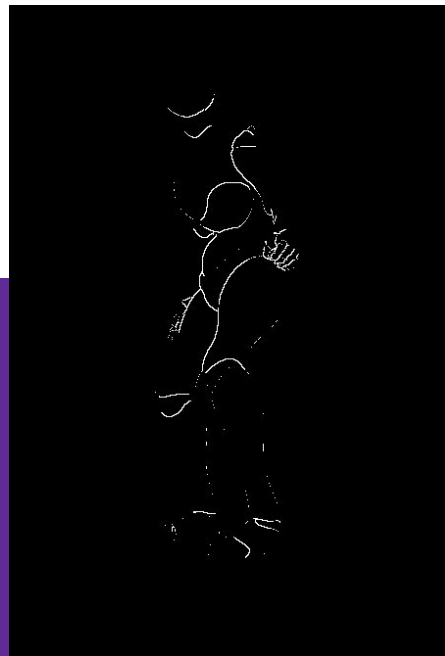
Detecting contours



mesh ID



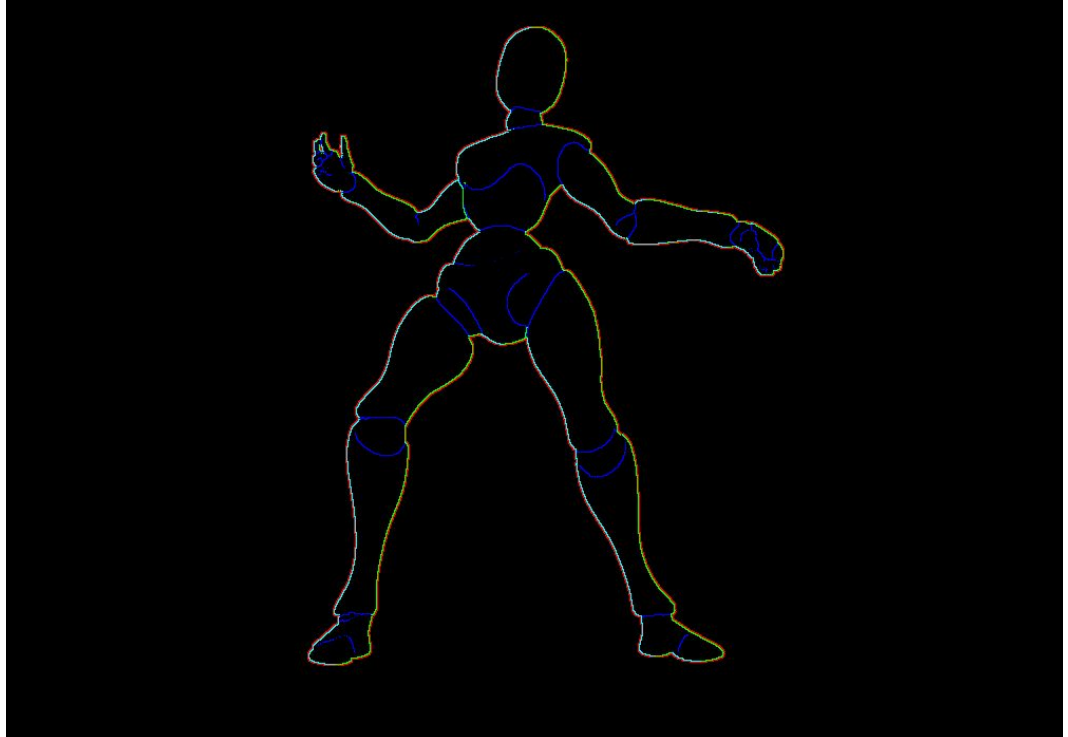
depth



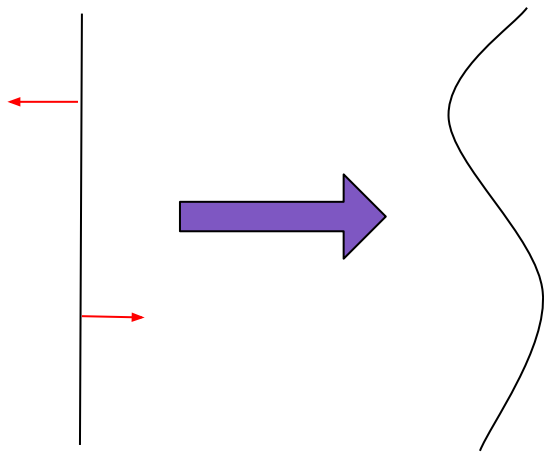
normal

Detecting contours

Techniques are
complementary



Adding randomness for sketchy effect

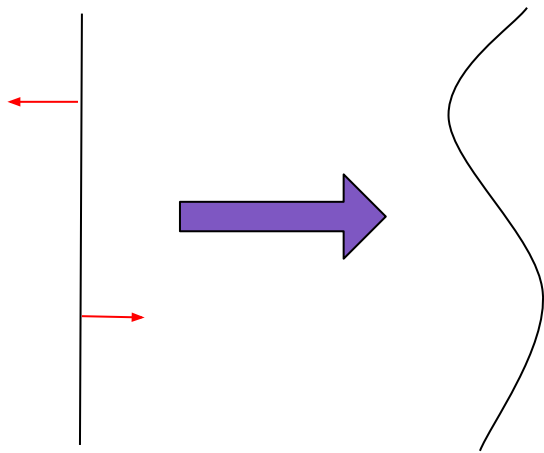


In the post-processing fragment shader :

```
newTexCoord = texCoord +  
noise;
```

```
fragColor = texture(myTexture,  
newTexCoord);
```

Adding randomness for sketchy effect



In the post-processing fragment shader :

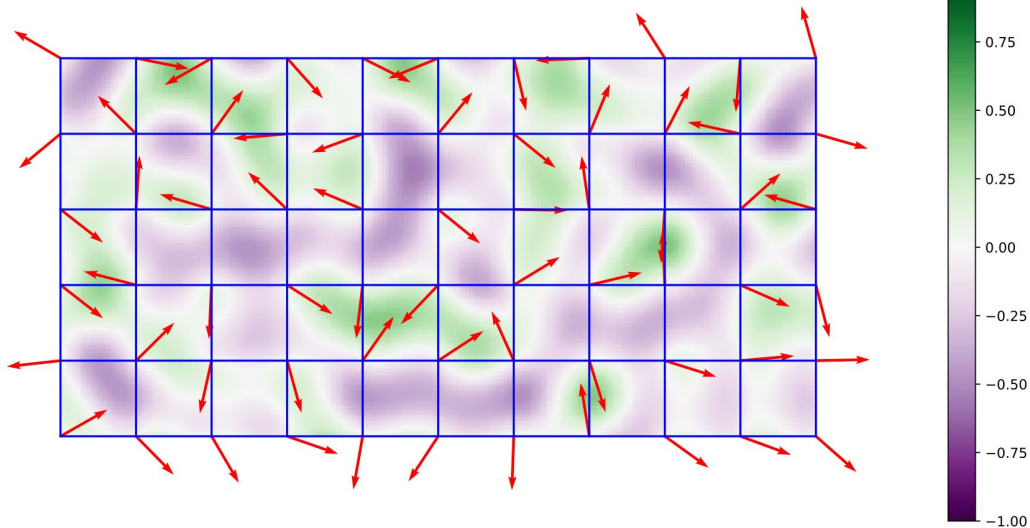
```
newTexCoord = texCoord +  
noise;
```

```
fragColor = texture(myTexture,  
newTexCoord);
```

Need continuous and “deterministic” noise
(all is done in parallel)

Perlin noise

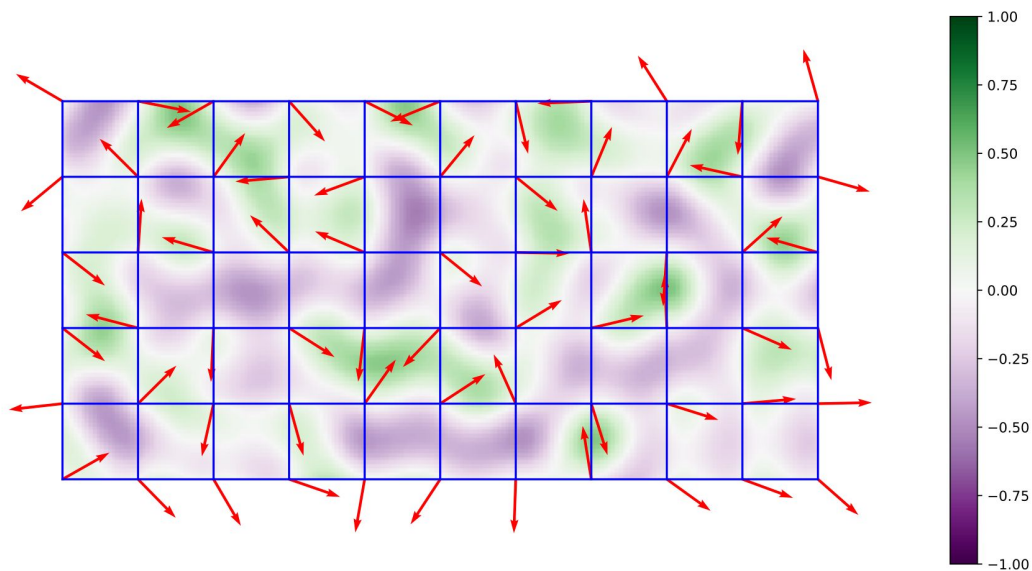
input : 2D, output : 1D



Source image : wikipedia.

- compute random gradient 2D vectors for grid nodes
- For each point : compute the dot product with 4 nearest grid nodes
- interpolate the 4 values weighted by the distance to the node

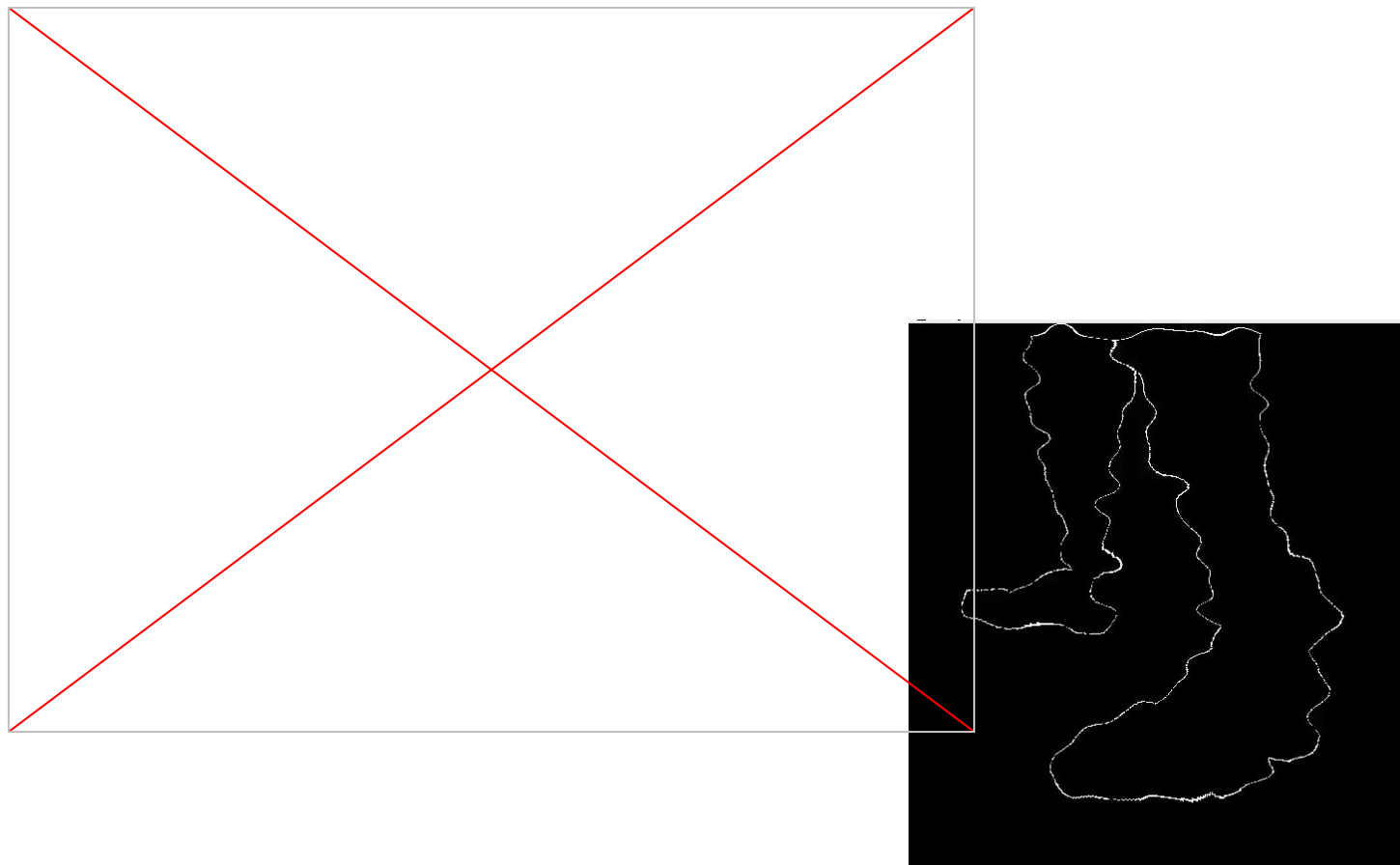
My Perlin noise concretely



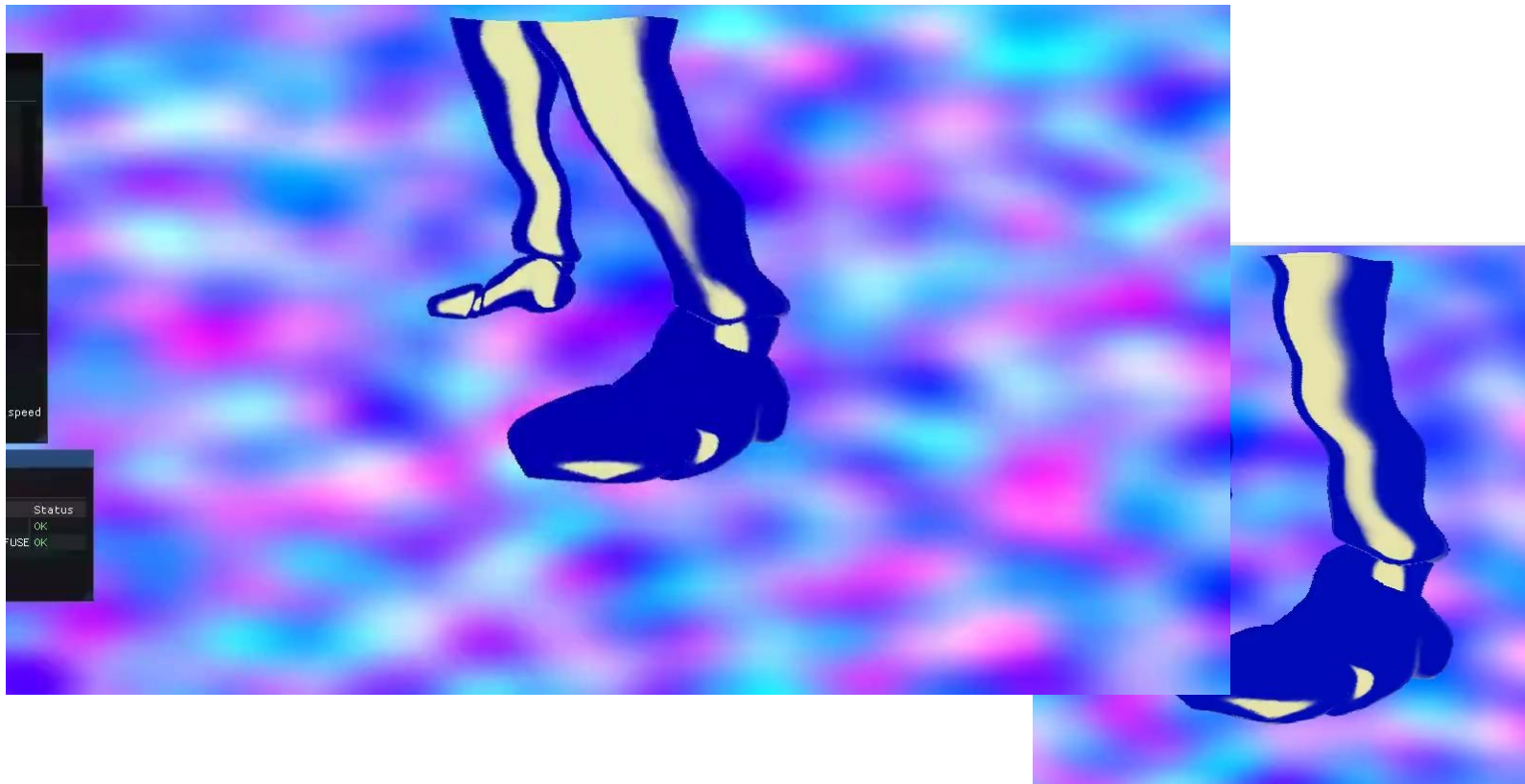
Source image : wikipedia.

- computed in a shader
- pseudo-random with same seed for all fragments
- input : 3D (x, y and time)
- output : 2D (texCoord displacement)

Result

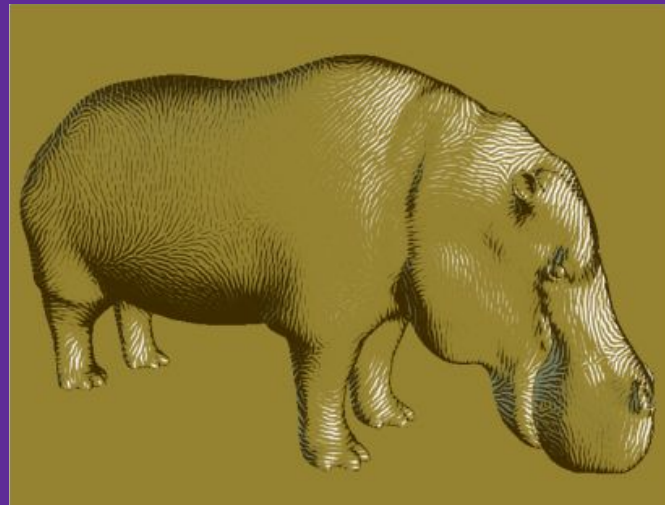


Cool bonus result



NPR shading

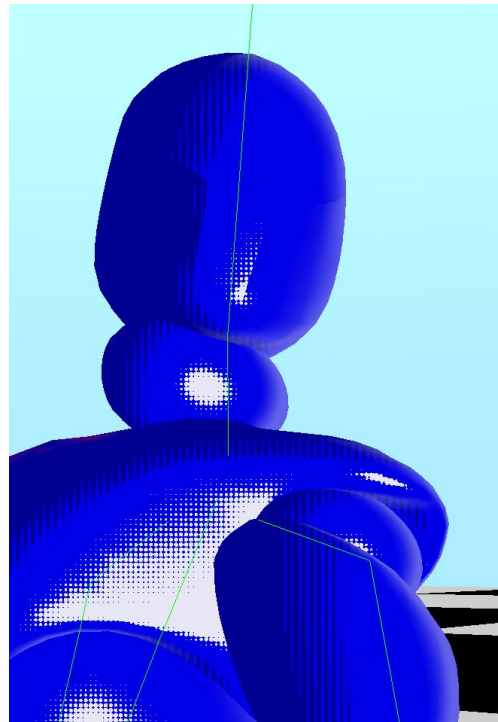
NPR = non-photorealistic
rendering



Source image : Breslav, Simon, et al. "Dynamic 2D patterns for shading 3D scenes."
ACM Transactions on Graphics (TOG) 26.3 (2007): 20-es.

Toon shading

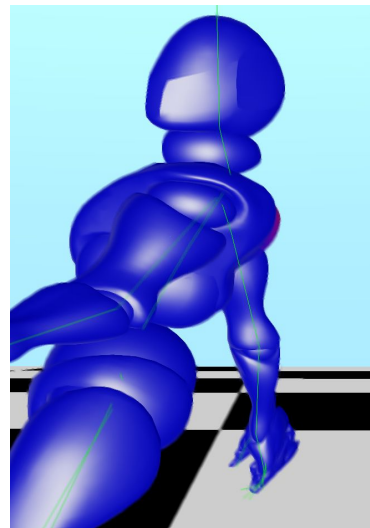
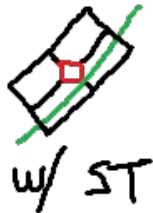
- Implemented primitives (SDFs)
- Smoothstep (sinusoidal)
- Activate with the right light contribution :
 - Ambient term : Cross-hatching
 - Diffuse term : Nothing
 - Specular term : Halftoning



Kuwahara anisotropic filter

3 render passes :

- Calculate the “structure” tensor for each pixel.
 - Input : Screen | Output : Structure tensor
- Smoothen the structure tensor image.
 - Input : Structure tensor | Output : Structure tensor
- Kuwahara filter with the modified pixel sampling method.
 - Input : Screen & structure tensor | Output : Screen

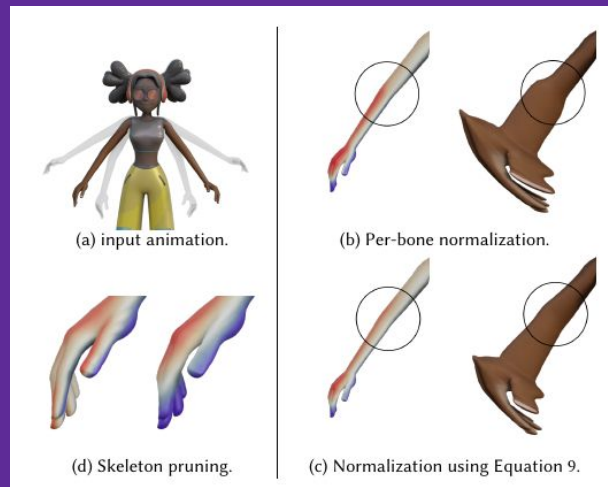
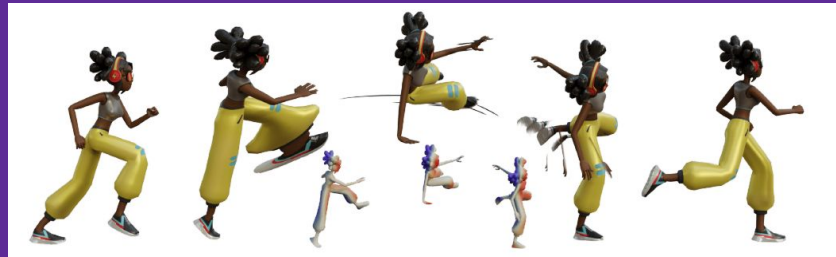


My result



Reference

Smear frame generation



Source image : Basset, Jean, Pierre Bénard, and Pascal Barla. "Smear: Stylized motion exaggeration with art-direction." *ACM SIGGRAPH 2024 Conference Papers*. 2024.

Motion offsets

- Indicate which parts of the model lead in front / trail behind
- Calculated for the whole animation beforehand

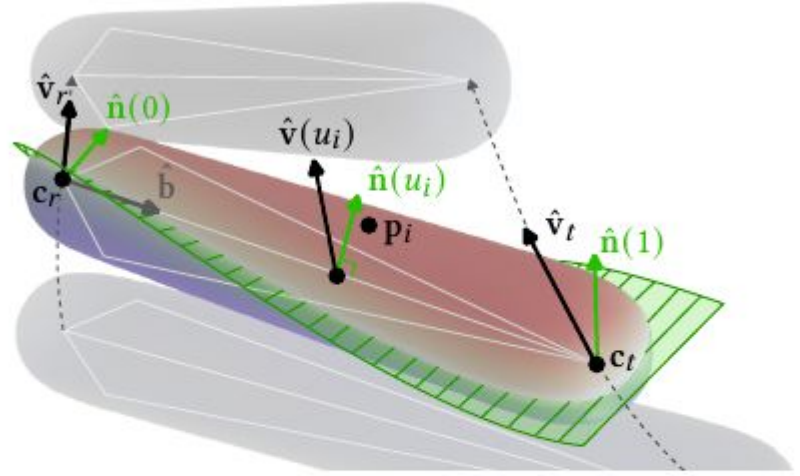
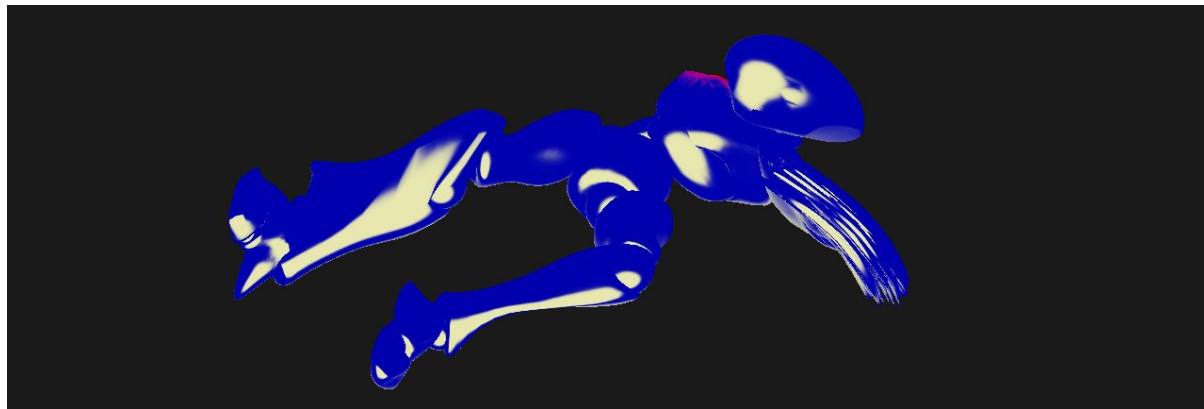
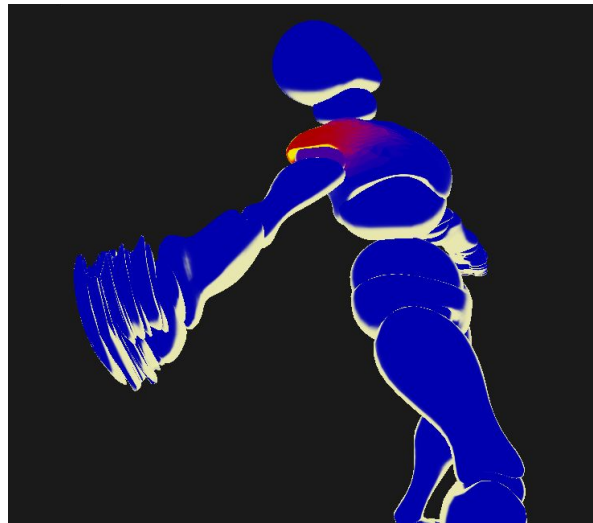


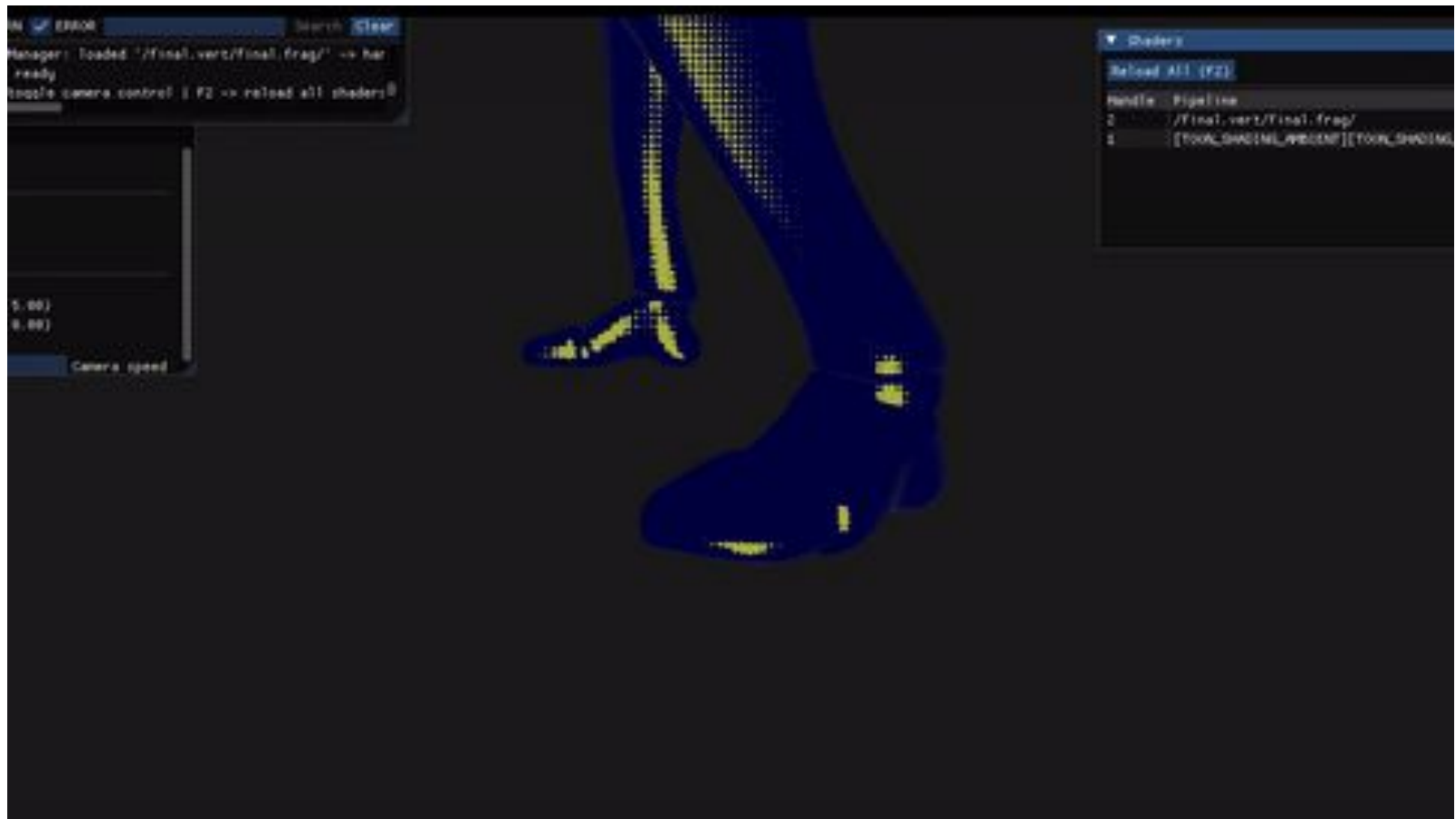
Fig. 5. Mathematical notations for a single bone. For every vertex i of the mesh with position $\mathbf{p}_i$, we compute the normal $\hat{\mathbf{n}}(u_i)$ of the ribbon (in green) which is orthogonal to the bone axis $\mathbf{b} = (\mathbf{c}_t - \mathbf{c}_r)$ and is as aligned as possible to the direction of motion $\hat{\mathbf{v}}(u_i)$ interpolated from the normalized velocities $\hat{\mathbf{v}}_r$ and $\hat{\mathbf{v}}_t$ at root and tip locations $\mathbf{c}_r$ and $\mathbf{c}_t$.

$$\bar{\delta}_i = w_{\text{coll.}}(u_i) \frac{\delta_i}{\max_j |\delta_j|} \quad \text{with} \quad \delta_i = (\mathbf{p}_i - \mathbf{c}_r) \cdot \hat{\mathbf{n}}(u_i), \quad (6)$$

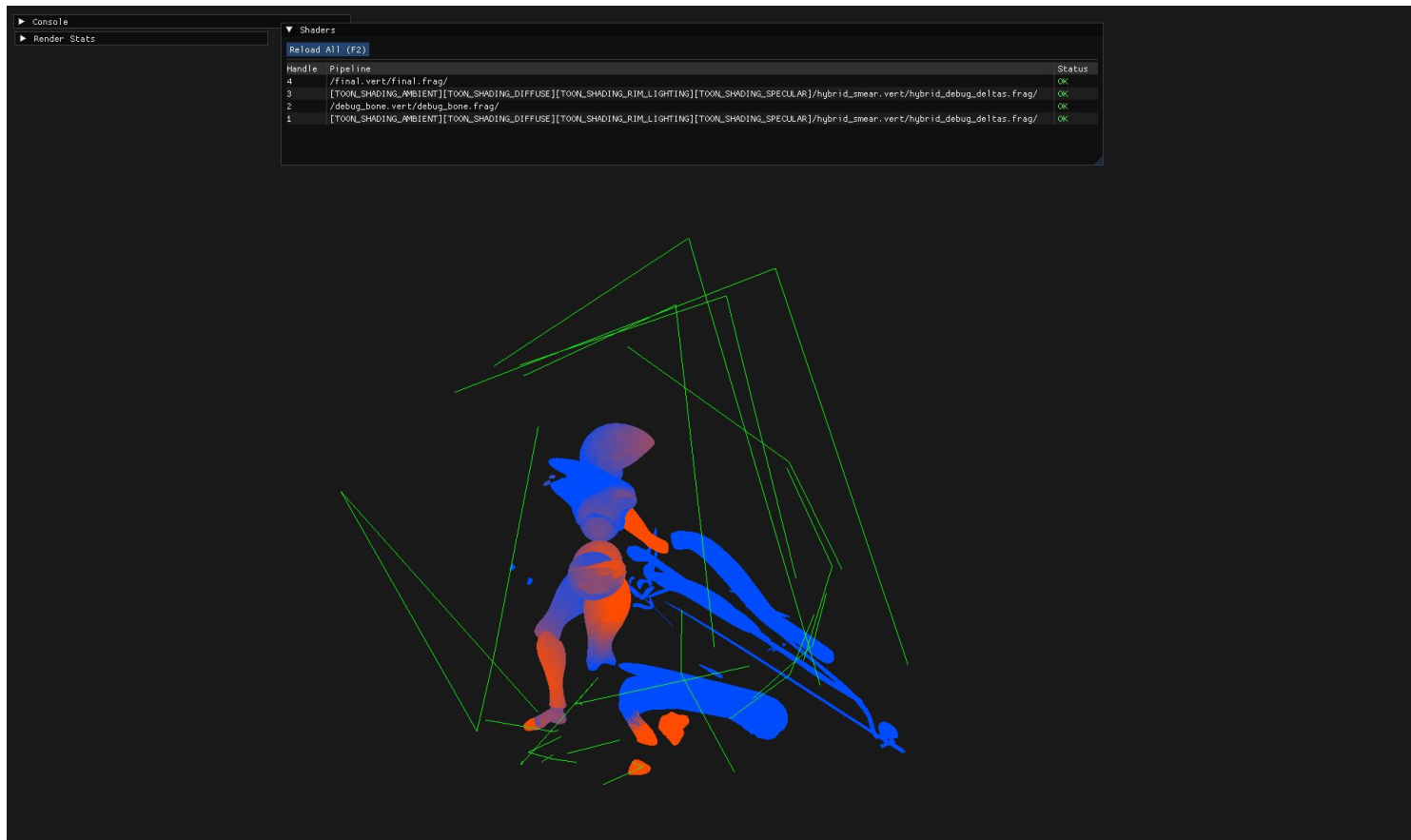
Elongated in-betweens

- Use motion offsets to move each vertex along a Catmull-Rom spline between animation frames

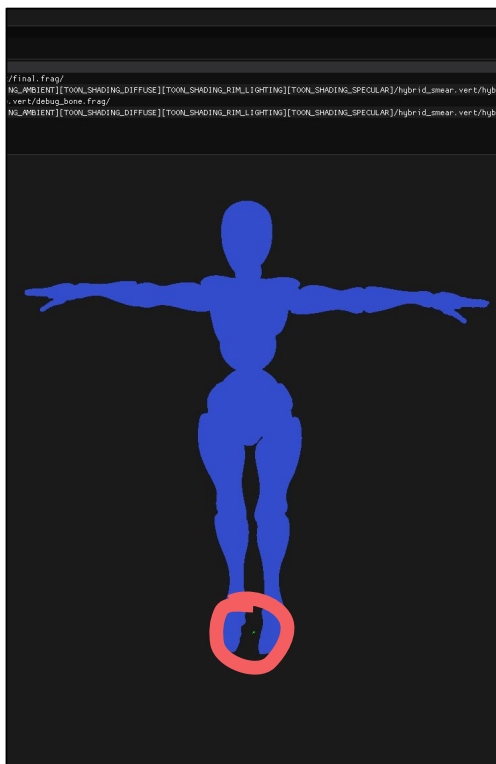




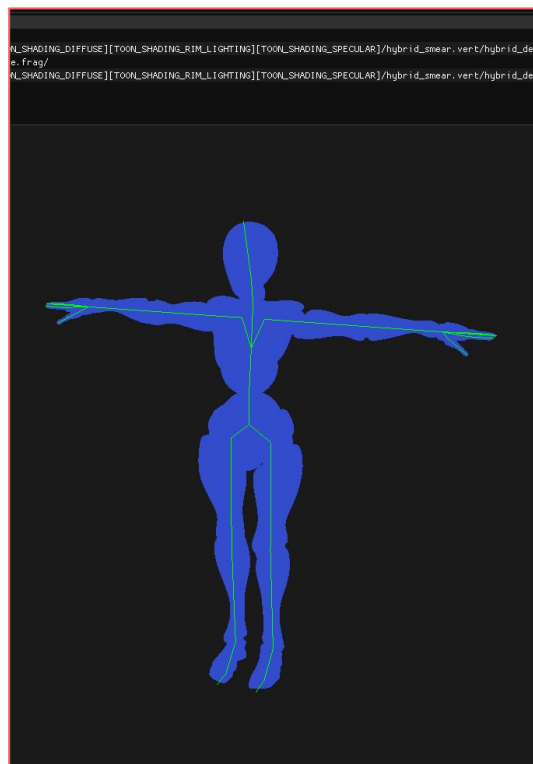
Previous attempts



Previous attempts

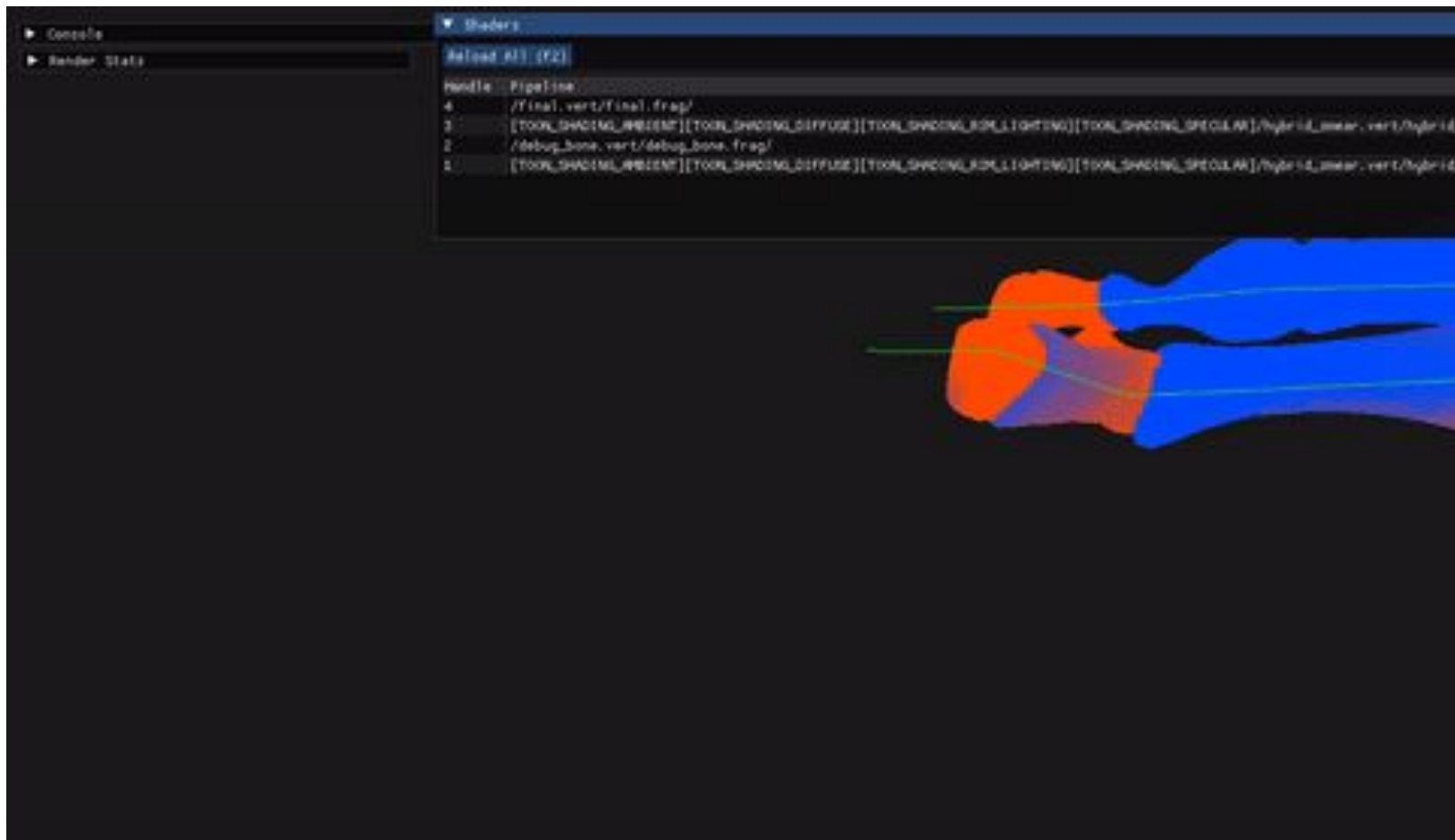


with offset



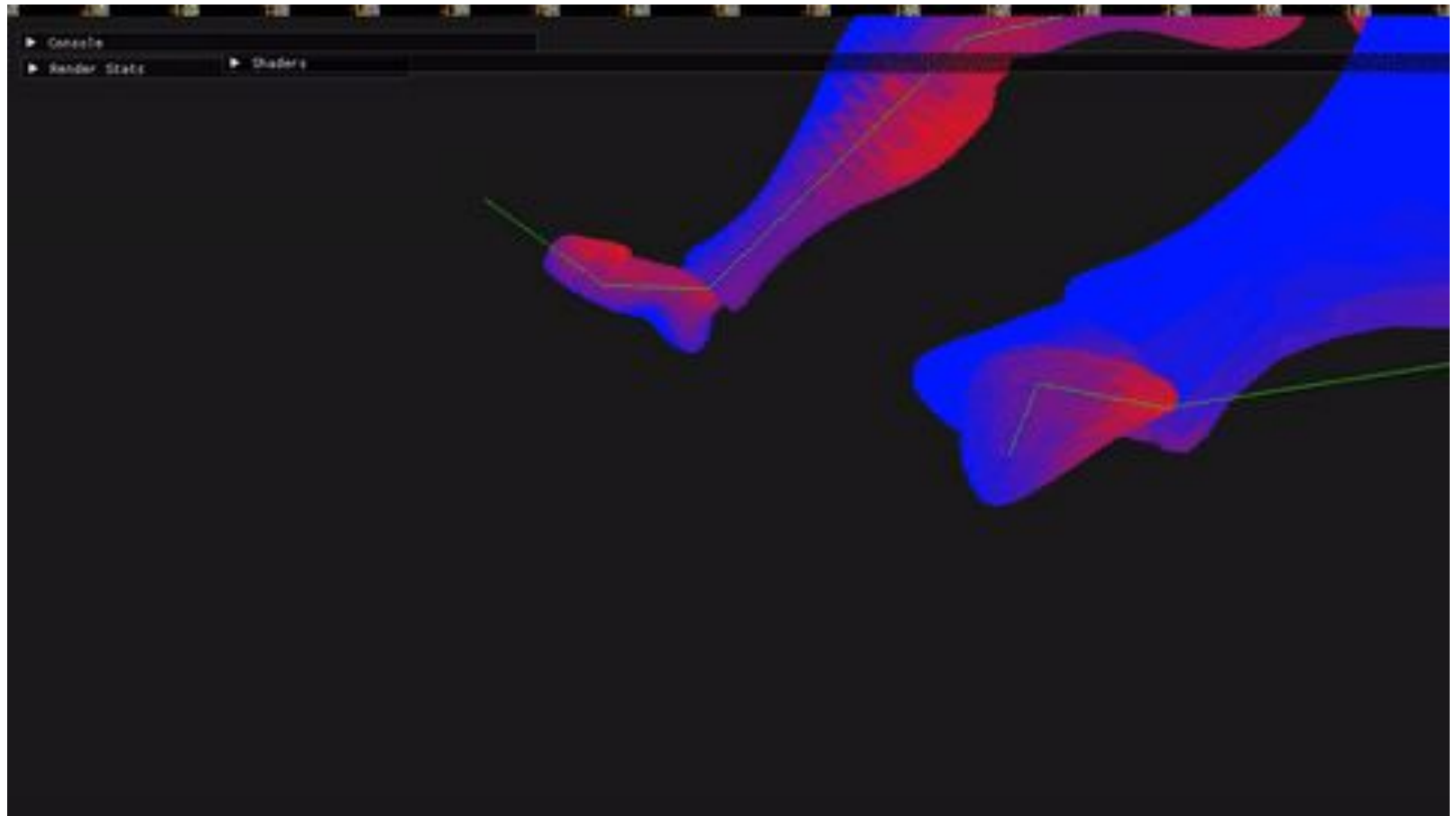
without offset

Previous attempts



Previous attempts

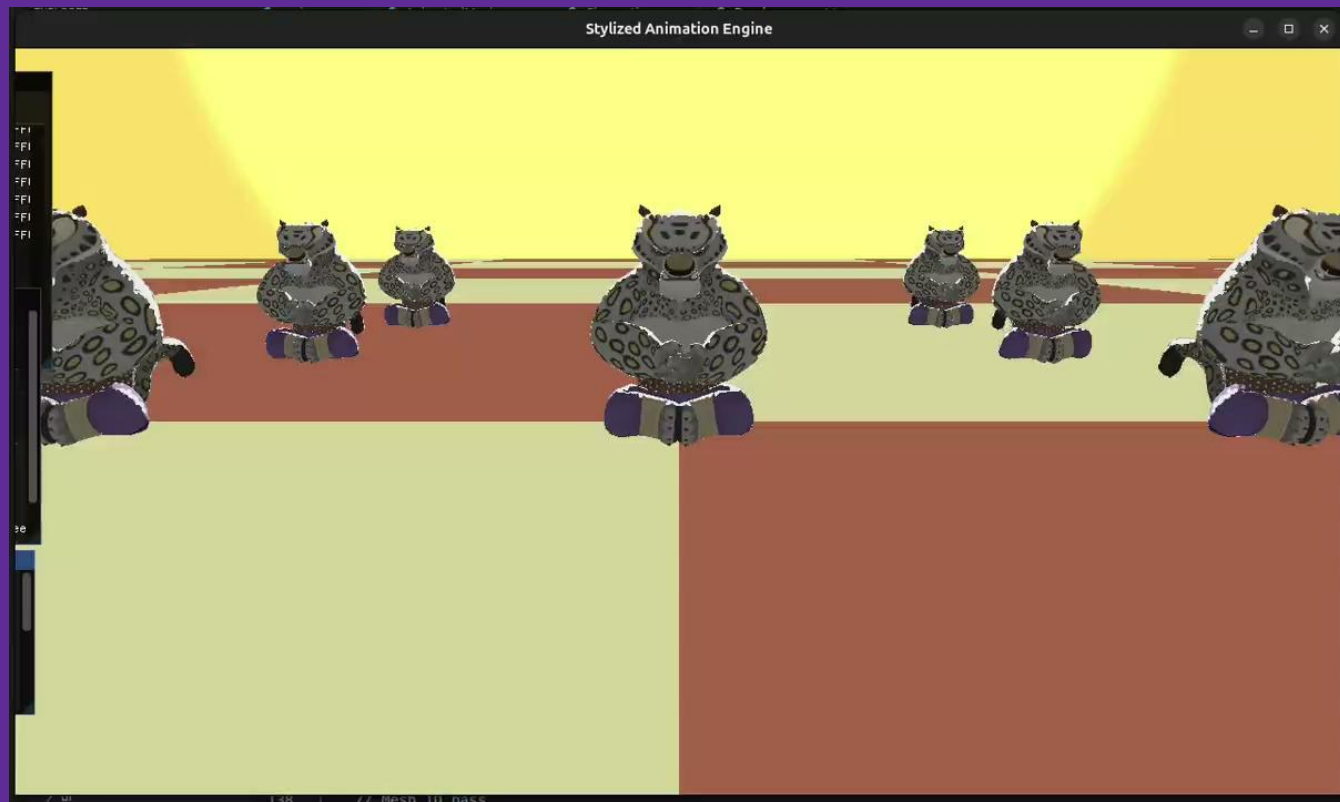
Problem: mismatch between world and bone space!



Final result

Our wonderful final scene

combining our works





Perspectives

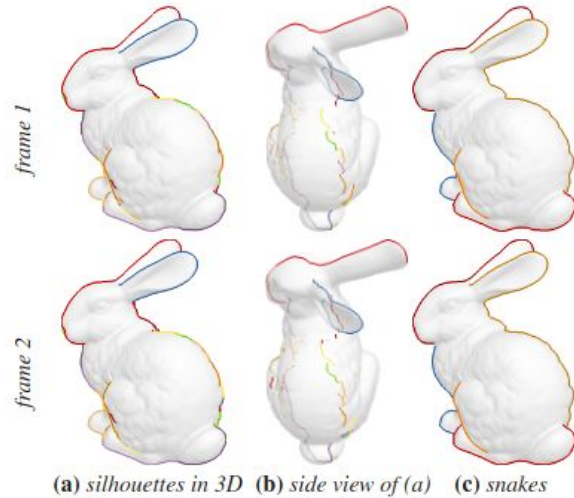
future work

Smear frames

- implement other stylization effects (multiple in-betweens and motion lines)
- propose a new solution for the self-rotation motion problem for articulated characters
 - For single rigid objects, a solution was already proposed in:

Tremolieres, Lou, et al. "Trajectory-aware Smears for Stylized 3D Animations." *Proceedings of the 2025 18th ACM SIGGRAPH Conference on Motion, Interaction, and Games*. 2025.

Contours : compute snakes to use brush textures



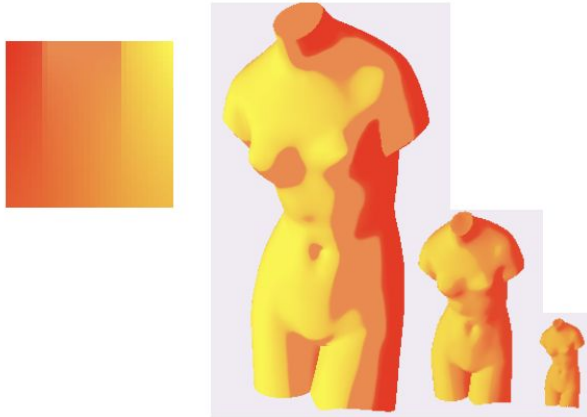
Source image : Pierre B nard, Lu Jingwan, Forrester Cole, Adam Finkelstein, Jo lle Thollot. Active Strokes: Coherent Line Stylization for Animated 3D Models.



Source image : MARKOSIAN, L., KOWALSKI, M. A., TRYCHIN, S. J., BOUR-DEV, L. D., GOLDSTEIN, D., AND HUGHES, J. F. 1997. Real-time nonphotorealistic rendering.

NPR shading

- Gloom effect
- Extended-Toon shading



User interface

better UI to

- change the final render pass' shaders
- pass uniform to shaders



Source image : Basset, Jean, Pierre B  nard, and Pascal Barla. "Smear: Stylized motion exaggeration with art-direction." *ACM SIGGRAPH 2024 Conference Papers*. 2024.